

Exact Structures and Maximal Canonically Jordan Recoverable Subcategories for Modules over Type A Algebras

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Question : Can a representation be recovered from the Jordan forms of its nilpotent endomorphisms?

In recent work, **Garver–Patrias–Thomas** (2018–2022) introduced and developed the notions of *Jordan recoverability (JR)* and *canonical Jordan recoverability (CJR)*. These ideas were later expanded by **Dequêne** (2022–2024).

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- $\text{rep}(Q)$ denotes the category of finite-dimensional left KQ -modules.
- $\mathcal{A}$ will denote a **hereditary abelian category**.
- All subcategories are assumed **full, closed under finite direct sums, and under direct summands**.

Jordan Form for Representations

Let $E = (E_q, E_\alpha) \in \text{rep} Q$ and $N = (N_q)_{q \in Q_0} \in \text{NEnd}(E)$. Each N_q is a nilpotent endomorphism of E_q .

Definition

The **Jordan form** of N_q is the partition $JF(N_q) \vdash \dim(E_q)$ recording its Jordan block sizes. Define the tuple

$$JF(N) = (JF(N_q))_{q \in Q_0},$$

which describes the block structure of N at each vertex.

The Generic Jordan Form Invariant

Fix $E \in \text{rep} Q$. For $N = (N_q)_{q \in Q_0} \in \text{NEnd}(E)$, set

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Definition [GPT23]

The **generic Jordan form** of E is the maximal value (with respect to dominance order):

$$\text{GenJF}(E) = \max_{N \in \text{NEnd}(E)} JF(N).$$

Question : Is this invariant complete?

Jordan Recoverability

Definition [GPT23]

Let $\mathcal{C} \subseteq \text{rep} Q$ be a subcategory. We say that $\mathcal{C}$ is **Jordan recoverable (JR)** if

$$E \simeq F \iff \text{GenJF}(E) = \text{GenJF}(F) \quad \text{for all } E, F \in \mathcal{C}.$$

Remark

The generic Jordan form GenJF is a *complete invariant* on $\mathcal{C}$.

Canonical Jordan Recoverability (CJR)

Definition [GPT23]

Let $\mathcal{C} \subseteq \text{rep} Q$. We say $\mathcal{C}$ is **canonically Jordan recoverable (CJR)** if for every $X \in \mathcal{C}$ there exists a dense open set (in the Zariski sense) $\Omega \subseteq \text{rep}(Q, \text{GenJF}(X))$ such that for all $Y \in \Omega$, $Y \simeq X$.

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Maximal CJR subcategory

A CJR subcategory $\mathcal{C}$ is **maximal** if it is not properly contained in any larger CJR subcategory of $\text{rep} Q$, i.e.

$$\forall \mathcal{D} \supsetneq \mathcal{C}, \quad \mathcal{D} \text{ is not CJR.}$$

Example on A_2 : JR but not CJR

Assume $Q : 1 \rightarrow 2$ and let $\mathcal{C} = \text{add}(S_1, S_2) \subseteq \text{rep}(Q)$. Every $E \in \mathcal{C}$ has the form

$$E = K^a \xrightarrow{0} K^b.$$

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Generic Jordan form

$$\text{GenJF}(E) = ((a), (b)).$$

Hence $\mathcal{C}$ is **JR**: $E \simeq F \iff \text{GenJF}(E) = \text{GenJF}(F)$ in $\mathcal{C}$.

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Open sets with generic Jordan form $(1, 1)$.

Consider $S_1 \oplus S_2 \cong K \xrightarrow{0} K$. Any dense open $U \subset \text{rep}_K(Q, (1, 1))$ contains representations $K \xrightarrow{\lambda} K \cong P_1$ with $\lambda \neq 0$.

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We have $\text{GenJF}(S_1 \oplus S_2) = ((1), (1)) = \text{GenJF}(P_1)$ with $P_1 \in U$, but $S_1 \oplus S_2 \not\cong P_1$. Hence $\mathcal{C}$ is **not CJR**.

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Theorem [Deq23a]

A subcategory $\mathcal{C} \subseteq \text{rep}(Q)$ is canonically Jordan recoverable if and only if every non-split short exact sequence

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

with $X, Z \in \mathcal{C}$, $Y \notin \text{ind}(\text{rep } Q)$.

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$$\begin{array}{ccccccc} 0 & \longrightarrow & X & \xrightarrow{f} & Y & \twoheadrightarrow & Z \longrightarrow 0 \\ & & h \downarrow & & \downarrow & & \parallel \\ 0 & \longrightarrow & D & \longrightarrow & PO & \twoheadrightarrow & Z \longrightarrow 0 \end{array}$$

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(ES3-II) Pullbacks

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Example of Exact Structures

$\mathcal{E}_{\max}$: the *maximal exact structure*, containing all short exact sequences.

$\mathcal{E}_{\min}$: the *minimal exact structure*, containing only the split short exact sequences.

Short Exact Sequences in Type A_n

Theorem

Let Q be an A_n type quiver. For a non-split exact sequence

$$0 \longrightarrow D \longrightarrow E \longrightarrow F \longrightarrow 0, \quad D, F \in \text{ind}(\text{rep } Q),$$

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- (a) $E \in \text{ind}(\text{rep } Q)$,*
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- (b) $E \cong E_1 \oplus E_2$ with $E_1, E_2 \in \text{ind}(\text{rep } Q)$.*

In case (b) we call it a **diamond exact sequence**.

The Diamond Exact Structure $\mathcal{E}_\diamond$

Definition

Let Q be an A_n type quiver. We define the collection of short exact sequences $\mathcal{E}_\diamond$ given by the additive bifunctor $\mathcal{E}_\diamond(-, -)$, uniquely determined by

$$\mathcal{E}_\diamond(N, M) = \{ \eta \in \text{Ext}^1(N, M) \mid \eta \text{ splits or diamond exact sequence} \}$$

for $M, N \in \text{ind}(\text{rep } Q)$ indecomposable.

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Proposition [Brüstle–Hanson–R.–Schiffler, 2024]

Let $n \in \mathbb{N}^*$ and Q be an A_n type quiver. Then $\mathcal{E}_\diamond$ is an exact structure on $\mathcal{A} = \text{rep}(Q)$.

Restatement of Theorem on CJR subcategories

Theorem

Let Q be an A_n type quiver. A subcategory $\mathcal{C} \subseteq \text{rep}(Q)$ is canonically Jordan recoverable if and only if $\mathcal{C}$ is $\mathcal{E}_\diamond$ -adapted.

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Remark. $\mathcal{C}$ is maximal canonically Jordan recoverable if any subcategory $\mathcal{D} \supset \mathcal{C}$ is not $\mathcal{E}_\diamond$ -adapted.

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The $\text{Gen}_{\mathcal{E}}$ and $\text{Sub}_{\mathcal{E}}$ Operators

Let $(\mathcal{A}, \mathcal{E})$ be an exact category.

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*For an object $M \in \mathcal{A}$, we set $\text{Gen}_{\mathcal{E}}(M) = \text{Gen}_{\mathcal{E}}(\text{add}(M))$,
 $\text{Sub}_{\mathcal{E}}(M) = \text{Sub}_{\mathcal{E}}(\text{add}(M)).$*

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Remark. $(\text{GS}_{\mathcal{E}}^i(\mathcal{C}))_{i \in \mathbb{N}}$ is an increasing sequence of subcategories of $\mathcal{A}$.

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The $\text{GS}_{\mathcal{E}}$ -operator is defined by

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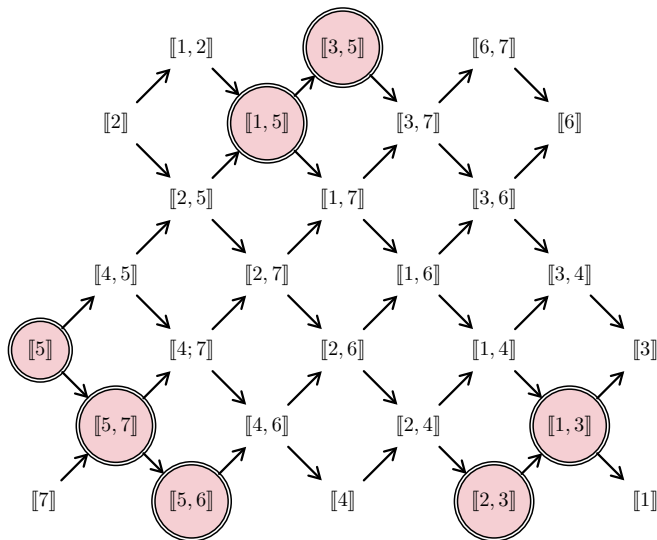
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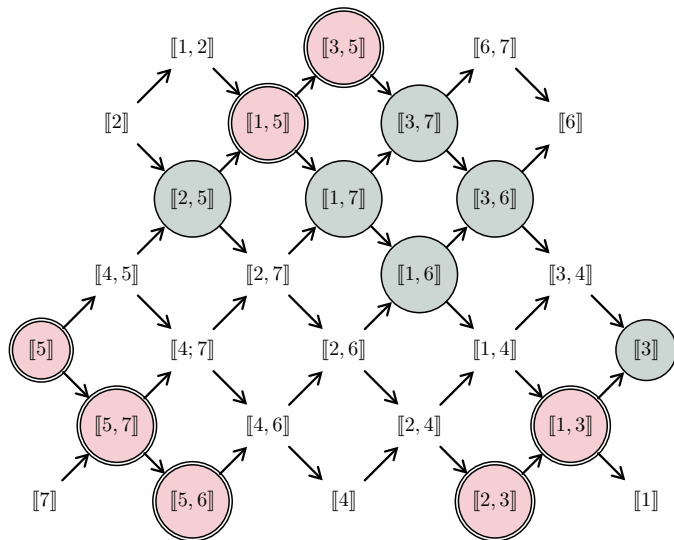
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$$\text{GS}_{\mathcal{E}_{\diamond}}^0(\textcircled{\textcircled{}}) = \text{add}(\textcircled{\textcircled{}})$$

$$Q = 1 \rightarrow 2 \leftarrow 3 \rightarrow 4 \rightarrow 5 \leftarrow 6 \rightarrow 7$$

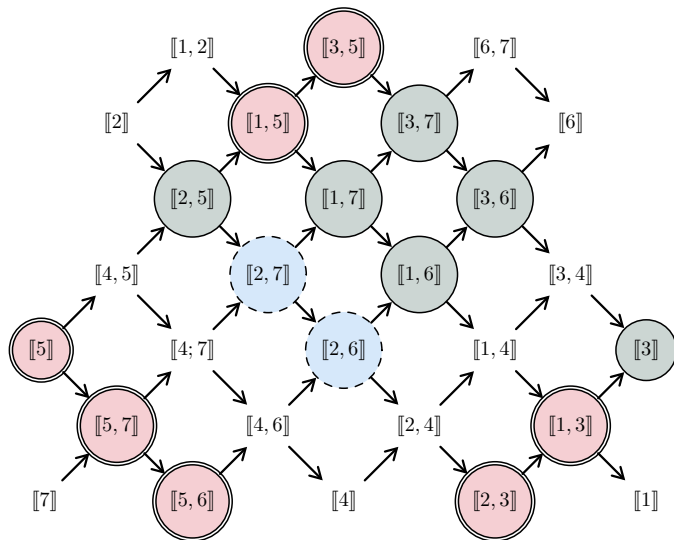


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$$\text{GS}_{\mathcal{E}_{\diamond}}(\textcircled{\textcircled{}}) = \text{GS}_{\mathcal{E}_{\diamond}}^2(\textcircled{\textcircled{}})$$

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Corollary 1. Let $T \in \mathcal{A}$ be a tilting object. Then $\mathrm{GS}_{\mathcal{E}}(T)$ is $\mathcal{E}$ -adapted and extension-closed.

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Corollary 2. Let $T \in \mathrm{rep}(A_n)$ be a tilting object. Then $\mathrm{GS}_{\mathcal{E}_{\diamond}}(T)$ is canonically Jordan recoverable.

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Question: Is $\mathrm{GS}_{\mathcal{E}_{\diamond}}(T)$ maximal canonically Jordan recoverable?

First Main Theorem

Theorem (Dequêne–R, 2025)

Let $T \in \mathcal{A}$ be a tilting object. Then $\mathrm{GS}_{\mathcal{E}}(T)$ is a maximal $\mathcal{E}$ -adapted, extension-closed subcategory of $\mathcal{A}$ containing T .

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Corollary

For any tilting object $T \in \mathcal{A}$,

$$\mathrm{GS}_{\mathcal{E}}(T) = \mathrm{GS}_{\mathcal{E}}^2(T).$$

First Main Theorem

Theorem (Dequêne–R, 2025)

Let $T \in \mathcal{A}$ be a tilting object. Then $\mathrm{GS}_{\mathcal{E}}(T)$ is a maximal $\mathcal{E}$ -adapted, extension-closed subcategory of $\mathcal{A}$ containing T .

In particular, every maximal canonically Jordan recoverable subcategory arises as $\mathrm{GS}_{\mathcal{E}_{\diamond}}(T)$ for some tilting object T .

Corollary

For any tilting object $T \in \mathcal{A}$,

$$\mathrm{GS}_{\mathcal{E}}(T) = \mathrm{GS}_{\mathcal{E}}^2(T).$$

How can we describe the equivalence relation :

$$T \sim_{\mathcal{E}} T' \iff \mathrm{GS}_{\mathcal{E}}(T) = \mathrm{GS}_{\mathcal{E}}(T')?$$

Plan

Classical and Relative Mutation

Definition

Let $T = U \oplus \overline{T}$ be a tilting object with U indecomposable.

A **mutation** of T at U is a tilting object

$$\mu_U(T) = U' \oplus \overline{T},$$

where U' is an indecomposable object obtained from U by an exchange sequence

$$\xi : 0 \rightarrow U \xrightarrow{f} B \xrightarrow{g} U' \rightarrow 0, \quad B \in \text{add}(\overline{T}),$$

where

f is a left minimal $\text{add}(\overline{T})$ -approximation of U ,

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We write $\mu_U^+(T)$ (resp. $\mu_U^-(T)$) for the *right* (resp. *left*) mutation.

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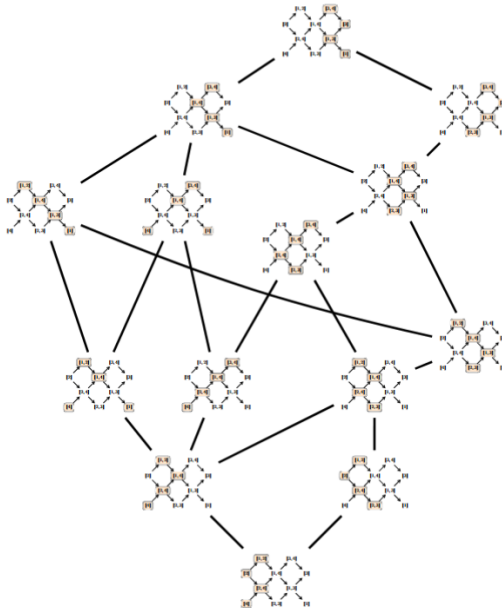
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Example : $Q = 1 \rightarrow 2 \leftarrow 3 \rightarrow 4$ with $\mathcal{E} = \mathcal{E}_{\max}$



$\mathcal{E}$ -Reachability

Definition

Let $T, T' \in \text{Tilt}(Q)$.

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T' is $\mathcal{E}$ -**reachable** from T if there exist indecomposables $(U_1, \dots, U_m)$ such that

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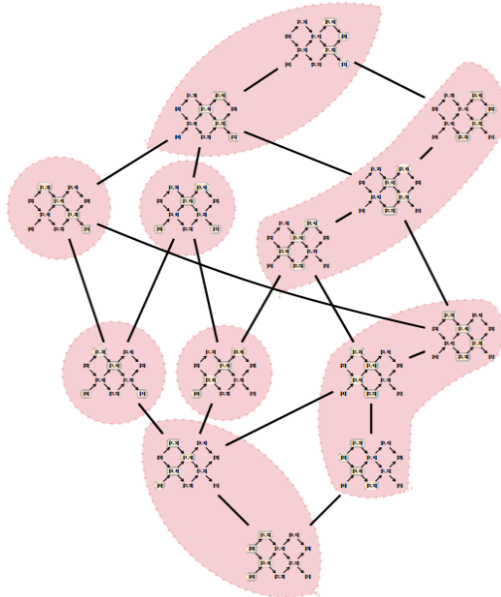
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$\mathcal{E}$ -reachability : $T \approx_{\mathcal{E}} T'$ forms an equivalence relation. We denote $[T]_{\approx_{\mathcal{E}}}$ for the equivalence class of T .

Example : $Q = 1 \rightarrow 2 \leftarrow 3 \rightarrow 4$ with $\mathcal{E} = \mathcal{E}_\diamond$



Equivalence of Tiltings

Proposition

Let $T, T' \in \text{Tilt}(Q)$. Then

$$T \sim_{\varepsilon} T' \iff T \approx_{\varepsilon} T'.$$

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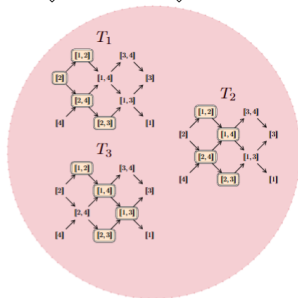
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$$[T_1]_{\approx_{\mathcal{E}_{\diamond}}} = [T_2]_{\approx_{\mathcal{E}_{\diamond}}} = [T_3]_{\approx_{\mathcal{E}_{\diamond}}}$$



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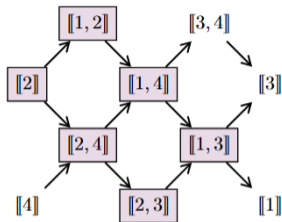
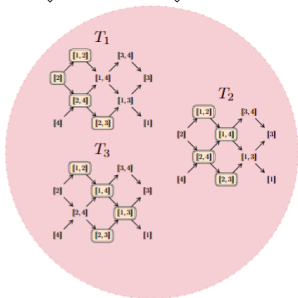
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$$[T_1]_{\approx_{\mathcal{E}_{\diamond}}} = [T_2]_{\approx_{\mathcal{E}_{\diamond}}} = [T_3]_{\approx_{\mathcal{E}_{\diamond}}} \quad \text{GS}_{\mathcal{E}_{\diamond}}(T_1) = \text{GS}_{\mathcal{E}_{\diamond}}(T_2) = \text{GS}_{\mathcal{E}_{\diamond}}(T_3)$$



Second Main Theorem

Theorem (Dequêne–R, 2025)

Let $T \in \text{Tilt}(Q)$. Then

$$\text{GS}_{\mathcal{E}}(T) = \text{add} \left(\bigoplus_{T' \in [T]_{\approx_{\mathcal{E}}}} T' \right).$$

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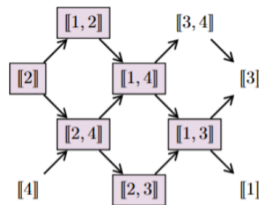
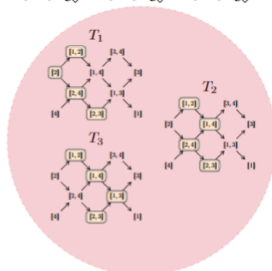
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Consequences of the Second Main Theorem

Corollary

Let Q be a quiver of type A_n .

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(a) There is a bijection

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Consequences of the Second Main Theorem

Corollary

Let Q be a quiver of type A_n .

(a) There is a bijection

$$\mathrm{Tilt}(Q)/\approx_{\mathcal{E}_\diamond} \longleftrightarrow \{\text{maximal CJR subcategories of } \mathrm{rep}(Q)\}.$$

(b) Let $\mathcal{C} \subseteq \mathrm{rep}(Q)$ be a maximal canonically Jordan recoverable subcategory. Then there exists $T \in \mathrm{Tilt}(Q)$ such that

$$\mathcal{C} = \mathrm{add} \left(\bigoplus_{T' \in [T]_{\approx_{\mathcal{E}_\diamond}}} T' \right).$$

Synthesis and Outlook

A unified picture of exact structures and recoverability

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Thank you for listening.

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Questions?

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